

Weak Compactness of State Spaces in $L^1([0, 1], X)$

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Status. `full_solution_likely_valid` for Problem 3.18 of arXiv:2509.12780, under the same separable-dual hypotheses used in the source paper’s Theorems 2.3 and 2.5. Problem 3.19, the nonseparable-dual variant, is not claimed.

1 Source Problem

The source is Soumitra Daptari and Saurabh Dwivedi, “A study on state spaces in classical Banach spaces,” arXiv:2509.12780, to appear in Colloquium Mathematicum [1]. For $x \in S(X)$, the state space is

$$S_x = \{x^* \in S(X^*) : x^*(x) = 1\}.$$

Under the standing identification $L^1([0, 1], X)^* = L^\infty([0, 1], X^*)$ when X^* is separable, the source defines S_f analogously for $f \in S(L^1([0, 1], X))$.

Problem 3.18, on page 18, asks for the weak compactness analogue of Theorem 2.5:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \langle x_i, x \rangle = \lim_{n \rightarrow \infty} \sum_{i=1}^n \langle x_i, x \rangle \quad \forall x \in X, \quad \left\| \sum_{i=1}^n x_i \right\| \leq 1$$

Problem 3.18. *In Theorem 2.3, we have characterized smooth point in $L^1([0, 1], X)$. In Theorem 2.5, we characterized the norm compactness of S_f for $f \in S(L^1([0, 1], X))$. But a result similar to Theorem 2.5 for weak compactness is not known.*

Problem 3.19. *The conclusions of Theorem 2.3 and Theorem 2.5 are not known when X^* is not separable.*

Conflict of interest: The authors declare that there is no conflict of interest.

In local source coordinates this is `data/parsed/arxiv_sources/2509.12780/Banach-CM.tex`, lines 442–447.

2 Idea of the Proof

The norm-compactness theorem in the source proves that a nonsmooth fibre set of positive measure lets one choose many separated pointwise states. For weak compactness, pairwise norm separation is not enough: weakly compact sets can contain separated sequences. The extra ingredient is to alternate between two measurably selected pointwise states by Rademacher signs. After evaluation at one fixed vector of X , this produces the usual Rademacher sequence in L^∞ . The closed span of that sequence is isometric to ℓ_1 , so no subsequence is weakly Cauchy. Eberlein–Smulian then forbids weak compactness of S_f .

3 Theorem

Theorem 1. *Let X be a real or complex Banach space such that X^* is separable, and let μ be Lebesgue measure on $[0, 1]$. For $f \in S(L^1(\mu, X))$, regard*

$$S_f = \{h \in S(L^\infty(\mu, X^*)) : \int_0^1 h(t)(f(t)) dt = 1\}.$$

Then S_f is weakly compact in $L^\infty(\mu, X^)$ if and only if f is a smooth point of $L^1(\mu, X)$.*

Equivalently, by Theorem 2.3 of [1], S_f is weakly compact if and only if $f(t) \neq 0$ almost everywhere and

$$\frac{f(t)}{\|f(t)\|}$$

is a smooth point of X for almost every t .

Proof. If f is smooth, then S_f is a singleton, hence weakly compact. We prove the converse.

We shall use the following elementary observation repeatedly. If $h \in S_f$, then the equality

$$1 = \int_0^1 h(t)(f(t)) dt \leq \int_0^1 \|h(t)\| \|f(t)\| dt \leq \int_0^1 \|f(t)\| dt = 1$$

forces $h(t)(f(t)) = \|f(t)\|$ for almost every t , after taking real parts in the complex case. Thus, off the zero set of f , $h(t)$ is a pointwise state of $f(t)/\|f(t)\|$ almost everywhere.

First suppose that $Z = \{t : \|f(t)\| = 0\}$ has positive measure. Choose $h_0 \in S_f$, choose $y \in S(X)$, and choose $y^* \in S(X^*)$ with $y^*(y) = 1$. Since Z is a positive-measure non-atomic Lebesgue set, it supports Rademacher functions $r_n : Z \rightarrow \{-1, 1\}$ such that every finite sign pattern occurs on a subset of positive measure. Define

$$h_n(t) = \begin{cases} r_n(t)y^*, & t \in Z, \\ h_0(t), & t \notin Z. \end{cases}$$

Then $h_n \in S_f$ for every n . The bounded evaluation map

$$P : L^\infty(\mu, X^*) \longrightarrow L^\infty(Z), \quad P(H)(t) = H(t)(y),$$

sends h_n to r_n . If S_f were weakly compact, Eberlein–Smulian would give a weakly convergent subsequence of (h_n) , and hence a weakly convergent subsequence of (r_n) in $L^\infty(Z)$. This is impossible: for every subsequence (r_{n_j}) , the closed span of $\{r_{n_j}\}$ is isometric to ℓ_1 , since for finite scalars (a_j) ,

$$\left\| \sum_j a_j r_{n_j} \right\|_\infty = \sum_j |a_j|.$$

The unit vector basis of ℓ_1 has no weakly Cauchy subsequence: after passing to any subsequence, the bounded functional with alternating coordinates $1, -1, 1, -1, \dots$ prevents convergence. This contradiction shows that $\mu(Z) = 0$.

Now assume $f(t) \neq 0$ almost everywhere and suppose that f is not smooth in $L^1(\mu, X)$. By Theorem 2.3 of [1], the set

$$E = \left\{ t : \frac{f(t)}{\|f(t)\|} \text{ is not smooth in } X \right\}$$

has positive measure. Put $u_t = f(t)/\|f(t)\|$ on E . Since X^* is separable, X is separable and $S(X^*)$, with the norm topology, is Polish. For some $\delta > 0$, the set of $t \in E$ for which $\text{diam } S_{u_t} \geq \delta$ has positive measure. The measurable graph argument used in the source paper, together with von Neumann selection, gives measurable maps $\varphi, \psi : E_0 \rightarrow S(X^*)$ on a positive-measure subset $E_0 \subseteq E$ such that

$$\varphi(t), \psi(t) \in S_{u_t}, \quad \|\varphi(t) - \psi(t)\| \geq \delta$$

for all $t \in E_0$.

Let (x_m) be a countable dense subset of $S(X)$. After replacing E_0 by a positive-measure subset B , we may fix $x_0 = x_m$ such that

$$a(t) := (\varphi(t) - \psi(t))(x_0)$$

satisfies $|a(t)| \geq \delta/2$ on B . Let $r_n : B \rightarrow \{-1, 1\}$ be Rademacher functions on B , and let $h_0 \in S_f$. Define

$$h_n(t) = \begin{cases} \frac{1+r_n(t)}{2} \varphi(t) + \frac{1-r_n(t)}{2} \psi(t), & t \in B, \\ h_0(t), & t \notin B. \end{cases}$$

On B , each value is either $\varphi(t)$ or $\psi(t)$, hence is a state of u_t . Outside B , h_0 is already a state for f almost everywhere. Thus $h_n \in S_f$ for all n .

Define a bounded linear map

$$Q : L^\infty(\mu, X^*) \longrightarrow L^\infty(B), \quad Q(H)(t) = \frac{2H(t)(x_0)}{a(t)}.$$

It is bounded because $|a(t)| \geq \delta/2$. If $m(t) = (\varphi(t) + \psi(t))/2$ on B , then

$$Q(h_n)(t) = Q(m)(t) + r_n(t) \quad (t \in B).$$

Weak compactness of S_f would again imply, by Eberlein–Smulian and boundedness of Q , that some subsequence of (r_n) is weakly convergent in $L^\infty(B)$, contradicting the same ℓ_1 -basis argument. Hence f must be smooth. $\square$

4 Consequence for Problem 3.18

Combining the theorem above with Theorems 2.3 and 2.5 of [1] gives the exact weak compactness analogue requested in Problem 3.18:

$$S_f \text{ weakly compact} \iff S_f \text{ norm compact} \iff f \text{ is smooth.}$$

Equivalently, under the source hypotheses, this happens precisely when $f(t) \neq 0$ a.e. and $f(t)/\|f(t)\|$ is smooth in X a.e.

5 Verification and Novelty Check

The proof is purely functional-analytic; no computational verification is needed. The key points for human review are:

- the measurable selection of two uniformly separated fibre states on a positive-measure subset;
- the bounded evaluation map Q , whose denominator is bounded away from zero after choosing a fixed vector x_0 ;

- the use of Eberlein–Smulian and the isometric ℓ_1 span of Rademacher functions in L^∞ .

Bounded novelty checks were performed in the run indexes for 2509.12780, state spaces, weak compactness, $L^1([0, 1], X)$, and S_f . No prior packet for this problem was found. Web searches for the exact paper title and phrases such as “state spaces weak compactness L^1 ” and “ S_f weakly compact $L^1([0, 1], X)$ ” found the source arXiv record but no later answer.

6 Limitations

This packet solves Problem 3.18 in the separable-dual framework of the source paper. It does not address Problem 3.19, where X^* is not separable and the measurable-selection machinery used both here and in the source paper is not directly available.

References

- [1] S. Daptari and S. Dwivedi, “A study on state spaces in classical Banach spaces,” arXiv:2509.12780, 2025; to appear in *Colloquium Mathematicum*.
- [2] J. Diestel and J. J. Uhl, *Vector Measures*, Mathematical Surveys, No. 15, American Mathematical Society, 1977.
- [3] S. M. Srivastava, *A Course on Borel Sets*, Graduate Texts in Mathematics 180, Springer, 1998.